

When Do Uncertain Dates Change Retrieval? Exact Certificates for Temporal Evidence

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Abstract

Sources often date historical changes only to a month or year. Guessing exact dates can change the evidence retrieved to answer a question. We ask when more precise dates could change that selection. Each claim states a fact whose start date lies within a known range. Our method summarizes each claim with at most two events that can replace it. These summaries distinguish evidence that could apply from evidence that must apply somewhere in the requested period. An exact test decides whether every permitted timeline selects the same evidence. When selection changes, we construct two valid timelines with different excerpts. These decisions require one scan, while asking whether any chosen claim could enter the context is NP-hard. A source-only pipeline retrieves evidence for 3,178 historical questions from TimeQA. It certifies 245 of 309 contexts with modeled evidence as stable and verifies alternatives for the remaining 64. A lazy selector performs fewer than three temporal tests per modeled context on average. The resulting interface directs date verification toward uncertainty that changes the reader’s evidence.

1 Introduction

Historical questions often ask about a period, while sources give incomplete event dates. A source might report an appointment in September without identifying its day. Choosing the first or last day can change which evidence reaches a reader. Yet resolving every uncertain date wastes effort when those dates cannot change the selected context. We ask a practical question: *When does date uncertainty actually change retrieval?*

This problem arises before answer generation. Retrieval-augmented systems select passages before generating answers (Lewis et al., 2020). Suppose a source states that a director’s term ended in 2005. The unknown departure day does not affect a question about 2004. For a question about June

2005, it can change whether that tenure remains eligible. Even then, the uncertainty matters only if the excerpt reaches the reader’s selected context. The system needs a test that distinguishes these cases before requesting more precise dates.

Temporal memories already preserve histories and invalidate replaced facts (Rasmussen et al., 2025; Zerhoubi et al., 2026). Interval retrievers also handle fuzzy dates (Chen et al., 2026; Wang et al., 2026). We study which retrieval decisions can be settled before completing those dates.

Each claim receives a range containing its unknown start date. A directed decision specifies which other claims can replace it. We compile these inputs into at most two source witnesses per claim. The first determines when support becomes impossible. The second determines when guaranteed support can fail. They answer every point or period query without enumerating timelines. *Possible support* means that a claim applies somewhere in the requested period under at least one allowed timeline. *Guaranteed support* means that it applies somewhere in that period under every allowed timeline.

The resulting masks also decide whether uncertainty changes retrieval. Fix a relevance ranking and a budget of k claims. If its first k possible claims are guaranteed, every timeline returns the same context. Otherwise, we identify a selected claim whose membership changes across two explicit timelines. These timelines can be replayed through the retrieval pipeline. They identify selected support decisions that further date evidence must resolve.

Our contributions are threefold. First, we derive exact period-support formulas and a sharp two-witness representation. Compilation takes linear time in the modeled events and supplied replacement pairs. Second, we characterize context stability and construct witnesses when it fails. Date refinement preserves settled decisions, while

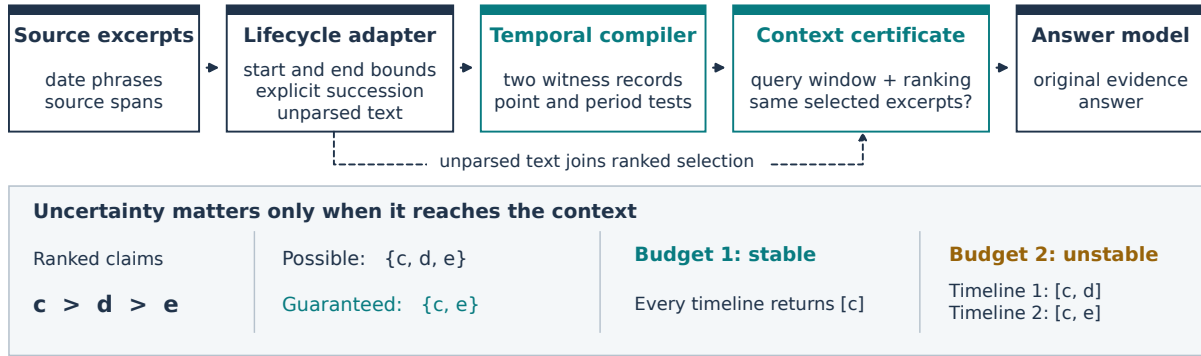


Figure 1: Source spans become dated claims with explicit replacement decisions. At most two witnesses per claim support exact temporal tests. A fixed ranking then yields a stable context or two admissible timelines with different contexts. Unparsed source text follows a shared fallback policy.

each changed replacement pair affects only its target. We also separate tractable stability certification from NP-hard possible context membership. Third, we integrate source extraction, title-aware retrieval, explicit endpoints, and a local answer model. Matched experiments separate evidence quality from date sensitivity on template and human-written questions. A lazy certificate scan examines only the candidates needed to fill the context.

2 From Sources to Temporal Evidence

An occurrence date says when a state began. A duration gives separate start and end dates. Table 1 shows how we represent both.

Occurrence precision. A day gives a point on the time axis. A month or year bounds an occurrence within that calendar period. Expressions such as *around 1713* require additional evidence before they supply finite bounds. We preserve the literal date expression and its source span. The query adapter applies the same convention to the visible question. A question about a whole year therefore remains a period query. Our predicate asks whether a claim applies *somewhere* within that period.

Source meaning	Representation
Started in	An unknown start between that month's first and last dates.
Served from	Separate uncertain start and end dates.
2001 to 2005	

Table 1: Date precision and state duration carry different information. Only an occurrence range bounds one unknown event date.

Replacement and coexistence. A directed replacement decision connects values of the same subject and role. For office succession, the organization and office define the slot; people supply its values. For affiliation, a new organization can coexist with an older affiliation. The Chicago Stadium passage in TimeQA, for example, gives overlapping tenancies for two teams (Chen et al., 2021). The source adapter records explicit succession separately from ordinary appointments or joins.

Explicit ends. A stated departure supplies an end event targeting its own claim. The compiler treats this hidden event as a direct replacement. Every permitted end must follow every permitted start: its lower bound exceeds the start's upper bound. This reduction preserves gaps between tenures. It also handles a departure without a named successor. Appendix E proves equivalence to the corresponding half-open lifetime.

Evidence units. A paragraph can contain several states, so temporal selection operates on source excerpts. Each modeled claim preserves exact character offsets and its supporting sentence for audit. Uncovered text remains available through a shared fallback policy. A loan's planned expiry never becomes an observed departure. An incomplete temporal parse retains the complete source text. Readers receive original excerpts, date bounds, and support tags. Source-derived subject and relation labels preserve context for short succession clauses.

Binding dates to claims. The source adapter requires an explicit local subject or a unique paragraph-level antecedent. It preserves abbreviation boundaries and checks which clause owns each date. It links a paragraph-level departure only

when the target is unique. Recognized bounded states with unresolved endings remain fallback text. These rules separate reliable date attachment from temporal inference.

Support tags. Modeled claims are guaranteed, unresolved, or unsupported for the requested period. Unresolved claims have possible support without guaranteed support. Unmodeled excerpts carry a separate *unknown* tag. Their fixed fallback eligibility lets us isolate uncertainty in modeled dates.

A source-grounded example. A Warburg Institute paragraph lists Henri Frankfort, Gertrud Bing, and Ernst Gombrich as successive directors (Chen et al., 2021). The paragraph dates their starts to 1949, 1955, and 1959, respectively. A deterministic clause adapter recovers these three starts and their two direct succession links from the paragraph. The passage separately dates Bing’s earlier organizational membership to 1922. That membership date does not start her directorship.

Consider the illustrative period January 1954 through June 1955. Frankfort has guaranteed support somewhere within that period. Bing has possible support because her appointment might precede July. Gombrich has no support within the period. Rank Frankfort first and Bing second. A one-item context always contains Frankfort. A two-item context changes when Bing’s start moves from January to December 1955. Both assignments satisfy the source’s year-level precision. The same possible pool thus yields different stability decisions at different budgets. The source is natural; the query, ranking, and assignments illustrate the theorem.

3 Dates as Constraints

Occurrence dates and replacement. Each claim c has text and an unknown occurrence date x_c . Its source bounds this date by two inclusive dates, $\ell_c \leq x_c \leq u_c$. Dates use a common real-valued time axis. These bounds describe uncertainty about an occurrence, rather than its duration. Each timeline independently chooses one date per range: $\Omega = \prod_c [\ell_c, u_c]$. A fixed text gate $G(d, c)$ records whether d can replace c . The gate need not be transitive. It excludes self-pairs: $G(c, c) = 0$. It must distinguish replacement from coexistence. The compiler preserves its decisions across date assignments.

For assignment x , claim c is active at date t when

$$A_x(c, t) = \mathbf{1}[x_c \leq t \wedge \nexists d : G(d, c), x_c < x_d \leq t]. \quad (1)$$

Equal dates do not establish replacement order. A question specifies an inclusive date window $Q = [a, b]$. Point questions set $a = b$. A claim supports the window when it is active somewhere inside it:

$$S_x(c, Q) = \mathbf{1}[\exists t \in Q : A_x(c, t) = 1]. \quad (2)$$

Possible support requires one allowed assignment with $S_x(c, Q) = 1$. Guaranteed support requires this condition in every allowed assignment. Write $P_Q(c)$ and $H_Q(c)$ for these two Boolean support decisions.

Two direct witnesses. One witness can force retirement despite date uncertainty. Another can permit retirement under a particular assignment. Their boundaries are

$$p_c = \min\{u_d : G(d, c), \ell_d > u_c\}, \quad (3)$$

$$r_c = \min\{\ell_d : G(d, c), u_d > \ell_c\}. \quad (4)$$

An empty minimum is $+\infty$. We store one witness for each finite boundary. The representation uses at most two witness records per claim. Two records are necessary in the worst case among direct witness subsets.

Theorem 1 (Exact window support). *For every claim and inclusive query window,*

$$P_Q(c) \iff \ell_c \leq b \wedge a < p_c, \quad (5)$$

$$H_Q(c) \iff u_c \leq b \wedge (a \leq \ell_c \vee a < r_c). \quad (6)$$

The first condition gives the smallest filter that never removes potentially supported evidence. The second gives the largest filter whose retained claims always have support. Each decision takes constant time after compilation. Appendix A gives the proofs.

Guaranteed support allows its supporting time to differ across timelines. For example, let $x_c \in [0, 10]$ and let a replacement start at time 5. Claim c always contributes somewhere in $[0, 10]$, either before the replacement or after its own later start. Yet no single point supports c across all timelines. A certainly active interval therefore cannot replace the window test.

Corollary 1 (Refinement and local influence). *Narrowing date bounds can only shrink possible support and expand guaranteed support. If two gates*

Input: claims and accepted pairs (d, c) .
Output: boundaries and their source witnesses.

1. Set $p_c = r_c = \infty$ for every claim.
2. For each accepted pair with $d \neq c$:
3. If $\ell_d > u_c$, minimize p_c using u_d .
4. If $u_d > \ell_c$, minimize r_c using ℓ_d .
5. Retain one supplying witness for each minimum.

Steps 3–4 execute inside the pair scan.

Figure 2: Linear compilation of the two support boundaries. The resulting records answer every query window.

differ on h directed pairs, either support mask changes for at most h claims. With fixed rankings, their filtered top- k contexts differ in at most $2 \min(k, h)$ items.

Each directed pair affects only its target claim’s certificates. When all date bounds are exact, both filters recover earliest direct replacement. The formulas therefore extend the same policy without choosing arbitrary dates.

4 Certifying the Reader’s Context

4.1 Compile once, query each claim in constant time

Figure 2 gives the complete scan. A stable source identifier breaks ties between witnesses.

Let n count modeled events, including explicit ends, and m count accepted pairs. Compilation takes $O(n + m)$ time and stores $O(n)$ certificate records. A support decision takes $O(1)$ time per claim. A complete mask takes $O(n)$ time. Pair discovery remains the source adapter’s responsibility. The reference complete scan makes $n(n - 1)$ gate calls. An index can replace this scan when it preserves every accepted pair.

4.2 An exact stability test

Let π_q be a complete relevance ranking for question q , including all tie breaks. For a set S , write $\text{Top}_k(S)$ for its first k ranked claims. The query’s possible and guaranteed sets are $P(Q)$ and $H(Q)$. A realized timeline x supplies a support set $S_x(Q)$. The reader’s selected claim context is

$$R_x(q) = \text{Top}_k(S_x(Q)). \quad (7)$$

Filtering preserves the ranking and occurs before the budget.

Theorem 2 (Exact context stability). *Every allowed timeline returns the same context exactly when*

$$\text{Top}_k(P(Q)) = \text{Top}_k(H(Q)). \quad (8)$$

Equivalently, every selected possible claim has guaranteed support.

Proof. Set $C = \text{Top}_k(P(Q))$. If $C \subseteq H(Q)$, every world retains C and selects the same context. Otherwise, choose $c \in C \setminus H(Q)$. Some world supports c , while another does not. At most $k - 1$ possible claims outrank c . Every world supporting c therefore selects it. The two worlds return different contexts. For $k = 0$, every context is empty. $\square$

The test uses two filtered rankings without enumerating timelines. Let F contain the fixed fallback excerpts. Apply the test to $P(Q) \cup F$ and $H(Q) \cup F$. Their inclusion remains constant while modeled dates vary. Claim selection precedes source rendering and any word-budget truncation. We report rendered-context agreement separately.

Choose a stable context budget. The same scan finds the largest stable prefix. Let r be the position of the first unresolved claim in the possible-support ranking. Every budget $k < r$ returns a fixed context, while every $k \geq r$ permits a context change. If no unresolved claim exists, every budget is stable. Each earlier modeled claim is guaranteed; earlier fallback excerpts stay eligible. The claim at position r supplies a counterexample for every budget that reaches it. The system can therefore report a stable prefix alongside the unresolved evidence requiring date refinement. This decision preserves the supplied relevance order.

Stop once the context is full. The implementation evaluates certificates lazily during selection. It scans ranked candidates until k possible excerpts fill the context. It checks guaranteed support only for selected modeled claims. Fixed fallback excerpts remain eligible without a temporal lookup. If selection examines j candidates, selection and certification take $O(j)$ time. Other dates need no evaluation for this decision.

4.3 Constructing and resolving an unstable context

Let $U = \text{Top}_k(P(Q) \cup F) \setminus (H(Q) \cup F)$. These claims have unresolved support and enter the pos-

sible context. Their status determines whether date uncertainty can change retrieval.

Theorem 3 (Constructive instability and refinement). *If $U \neq \emptyset$, we construct two complete date assignments with different contexts in $O(n)$ time. Here, n counts the event dates. Every date refinement that stabilizes the context makes each claim in U guaranteed or impossible.*

Each $c \in U$ has fewer than k possible claims above it. Any timeline supporting c therefore selects it. We place each accepted replacement of c no later than c or after a supporting query time. A second timeline places c after the query, or a direct witness by the query start. Date refinement can only remove possible claims. Thus a remaining possible pivot cannot leave the possible context through rank changes. It must become guaranteed. The assignments use only event bounds and the compact certificate. Let N count ranked excerpts, including fallback text. Full context replay takes $O(n + m + N)$ time with the accepted pair list. Appendix F gives the construction and proof.

A frontier claim can require a more precise start date or replacement date. The source witnesses locate the evidence needed for that decision. A system can request this evidence, narrow its date bounds, and compile a new certificate. New unresolved claims can enter after current pivots disappear. We recompute the frontier after each refinement.

4.4 A tractability boundary

Possible support concerns one claim at a time. Possible context membership also depends on which higher-ranked claims coexist. We ask whether a given claim enters the context under *any* allowed timeline.

Theorem 4 (Possible context membership). *When k is part of the input, deciding possible context membership is NP-hard.*

This holds even for a point query and a bipartite gate with one edge layer.

Proof. Take a Set Cover instance with elements E , sets $S_1, \dots, S_s$, and budget $1 \leq B \leq s$ (Karp, 1972). Each set becomes a claim v_j with start bounds $[1, 3]$. Create $B + 1$ copies of each element, all starting at time 0. Add a candidate c starting at time 0. A pair $v_j \rightarrow e$ targets every copy of each element in S_j . No other pairs are accepted. Rank

every set claim and element copy above c . Use point query 2 and context budget $B + 1$.

For a cover of at most B sets, start its set claims at time 1. Start the others at time 3. Every element copy retires before the query. At most B higher-ranked set claims remain active, so c enters the context. Conversely, a context containing c has at most B active higher-ranked claims. Its active set claims must cover every element. An uncovered element would leave $B + 1$ active copies above c and exclude it. Thus these claims define a cover of size at most B . The construction is polynomial because $B \leq s$. $\square$

Appendix F also gives a reduction where every start precedes or equals the query time. Our stability test instead needs one unresolved claim inside the possible prefix. Its rank guarantees selection whenever it has support. The test needs only individual support, even when shared events couple claim eligibility. This makes exact stability and its witnessing timelines tractable without computing every attainable context member.

4.5 What the witnesses explain

Each certificate stores a target’s bounds, both minima, and their supplying source identifiers. An exclusion caused by retirement points to its direct replacement witness. An exclusion caused by a late start points to the target’s own date bounds. This makes the temporal decision inspectable without reconstructing a cluster history.

The two boundaries answer different questions. For $c = [0, 2]$, take accepted witnesses $d = [1, 4]$ and $e = [3, 5]$. Witness d supplies $r_c = 1$ because it can follow an early start of c . Witness e supplies $p_c = 5$ because it must follow every start of c . Keeping only d incorrectly allows c at time 5. Keeping only e incorrectly guarantees c at time 2. Neither one-witness subset reproduces every query decision.

The same direct structure limits update propagation. Adding events whose earliest dates exceed the query end preserves that query’s earlier support decisions. Later ingestion can still correct a historical date when its event bounds reach the queried period. These properties depend on direct temporal support, rather than document arrival order.

5 Evaluation Protocol

We evaluate support decisions, context certificates, retrieved evidence, and generated answers. The

natural-source experiments use original TimeQA paragraphs (Chen et al., 2021). Source extraction receives article titles, paragraph text, identifiers, and paragraph order. Retrieval receives only the visible question. Question identifiers serve as opaque scoring keys. Answers and annotated spans enter scoring only.

Calibration and evaluation. A fixed hash selects 176 development questions from 50 articles for ranking calibration. Separate source inspections guide extraction rules and their semantic audit. The revised source adapter remains fixed throughout evaluation. We evaluate 2,348 template questions from 644 articles and 830 human-written questions from 251 articles. Both tracks search their complete source pools. These contain 32,332 and 33,493 paragraphs, respectively. The appendix records earlier inspected outcomes, calibration exclusions, and all input hashes.

Fixed ranking and matched controls. The selected ranking links the longest public article title appearing in the question. It removes matched title words before BM25 scoring (Robertson et al., 1995). We use $k_1 = 1.5$ and $b = 0.75$. Linked excerpts receive priority; all other corpus excerpts remain in the complete ranking. Six ranking variants compete on the pilot before confirmation. Every temporal policy shares source units, bounds, replacement pairs, ranking, and tie breaks. We compare earliest, midpoint, and latest date completion against possible and guaranteed support. The interval control tests start and explicit-end overlap, ignoring cross-claim replacement. The no-filter control retains all excerpts. Separate ranking controls use original BM25 and latest-year reranking within twenty lexical candidates. We also apply that recency rule to the selected title-aware ranking.

Evidence budgets and measurements. Each context holds five source excerpts and at most 768 source-text words. Words are separated by whitespace; source titles and temporal metadata add prompt text. Every method retains unmodeled text under the same fallback policy. We average each question’s fraction of annotated spans fully contained in its shown excerpts. A parent paragraph match alone earns no span credit. We report stability separately for contexts with modeled claims and contexts containing only fallback text. Every unstable context receives two complete date

Certificate cohort	Template	Human
Questions	2,348	830
Parsed query periods	2,296	813
Contexts with modeled evidence	243	66
Certified stable contexts	189	56
Unstable contexts	54	10
Stable within modeled group	77.78%	84.85%

Table 2: Stability for the frozen title-aware ranking. The modeled group contains parsed questions with at least one modeled excerpt in the possible context. All-fallback contexts have a separate denominator.

assignments with different selections. An independent evaluator replays those assignments. Confidence intervals use 2,000 paired article-cluster resamples, with seed 20261006.

Answer generation. We run Qwen2.5-3B-Instruct with official Q4_K_M weights and a pinned CPU runtime (Qwen Team, 2024). A fixed hash selects 60 human questions from 55 articles before generation. Ten methods yield 600 logical conditions. Identical prompts share one generation, with every question–method membership retained. Temperature is zero, seed is 1729, and the output limit is 96 tokens. The prompt requests all supported answers and cited excerpt numbers in JSON. Parsing follows the earlier frozen protocol. Invalid responses contribute empty answers and remain in every denominator. We score exact annotated answer sets and strict set F1. The release contains complete prompts, raw outputs, timing, and request hashes.

6 Results

6.1 Most selected contexts need no finer modeled dates

Table 2 reports exact context decisions under the frozen title-aware ranking. Among 243 template contexts containing modeled evidence, 189 remain unchanged across all permitted timelines. The corresponding human group has 56 stable contexts among 66. Their stability rates are 77.78% and 84.85%, respectively. The remaining contexts contain selected support decisions that date refinement must resolve.

The adapter extracts 493 development claims and 464 test claims. They occupy 1.49% and 1.36% of indexed source units, respectively. Explicit end events number 292 and 238. The source revision adds sixteen paragraph-level end links. All 27 source checks pass before evaluation. The eval-

Excerpt budget k	1	3	5	10	20
Template ($n = 243$)	90.9	79.4	77.8	77.0	76.5
Human ($n = 66$)	97.0	90.9	84.8	83.3	78.8

Table 3: Stable contexts (%) as the unit budget grows. Each row retains the same modeled cohort selected at $k = 5$. Selection precedes word truncation.

uation cases exercise uncertain starts and explicit ends. All three recovered succession pairs belong to calibration articles. A recorded model-assisted source audit accepts 54 of 60 sampled event interpretations. Appendix G reports component judgments and earlier source audits.

Parsed all-fallback contexts number 2,053 and 747 across the two tracks. Their evidence stays fixed under the shared fallback policy. We keep them outside the modeled-context stability rates.

6.2 The budget exposes consequential uncertainty

Table 3 varies only the excerpt budget on each fixed modeled cohort. At one excerpt, stability reaches 90.9% for template questions and 97.0% for human questions. Even at twenty excerpts, more than three quarters of these contexts remain stable. A system can retain a short stable context while exposing the extra evidence requiring date verification.

The lazy selector examines only the candidates needed to fill the context. On modeled template contexts, it examines 5.21 candidates and performs 2.63 temporal tests on average. The human averages are 5.11 candidates and 2.44 tests. The full masks would evaluate both support predicates for all 493 or 464 claims. All 3,109 parsed queries match the full-mask selections exactly. These counts isolate selection after the relevance ranking has been computed.

6.3 Unstable contexts return replayable alternatives

Every unstable context receives two complete assignments to the modeled event dates. Independent replay verifies all 128 assignments and all 64 changes in rendered source excerpts. Across both tracks, 41 exclusions retire the pivot by the query start. Twenty-three move its start beyond the query end. Sixty refinement frontiers contain one claim, and four contain two. The interface returns these current support decisions with their source spans.

Method	Template	Human
Original BM25	37.63	44.80
Original BM25 + latest year	53.27	53.11
Title-aware ranking	67.61	58.11
+ earliest completion	67.87	58.47
+ midpoint completion	67.76	58.35
+ latest completion	67.78	58.35
+ interval control	67.89	58.47
+ possible support	67.89	58.47
+ guaranteed support	67.76	58.35
Title-aware + latest year	66.55	64.26

Table 4: Mean question-level annotated span recall (%) within five excerpts and 768 source-text words. Temporal controls share the title-aware ranking. Latest-year controls change only ranking.

A month-level departure changes the context.

A human question about Helen Clark specifies the period 1989 through August 1989. The source dates her conservation tenure from August 1987 until January 1989. An end on January 1 removes that excerpt from the requested period. An end on January 31 retains it. Both dates respect the source’s month-level precision. The command-line interface reproduces both contexts from the frozen source records. More precise evidence needs to settle this boundary; the system does not require every historical date to become exact.

6.4 Stronger retrieval and exact certificates compose

Table 4 compares shown-span recall under identical source units and budgets. The pilot-selected title-aware ranking reaches 67.61% on template questions and 58.11% on human questions. It improves over original BM25 with recency by 14.34 and 5.00 percentage points. The paired intervals are [11.86, 16.77] and [0.69, 9.10] points.

Adding recency to the title-aware ranking reaches 64.26% on human questions. Its improvement over the original recency control is 11.14 points, with interval [7.58, 15.06]. Possible support and the interval control select identical excerpts on these tracks. The same evidence pipeline therefore combines improved retrieval with exact uncertainty certificates.

A secondary analysis applies the certificate to all six prespecified ranking variants. It reads no answer labels and preserves the frozen main ranking. Under the strongest human retrieval control, 102 of 120 modeled contexts remain stable. Across variants, mean temporal tests range from 2.34 to 2.44 per modeled human context. The appendix

Method	Primary	Diagnostic
Original BM25	16.67	9.09
Original BM25 + year	18.33	9.09
Title-aware ranking	21.67	9.09
+ earliest completion	20.00	18.18
+ midpoint completion	20.00	9.09
+ latest completion	20.00	9.09
+ interval control	20.00	9.09
+ possible support	20.00	9.09
+ guaranteed support	20.00	18.18
Title-aware + year	21.67	15.15

Table 5: Strict answer-set F1 (%). The primary sample has 60 questions. The diagnostic retains eleven previously selected context-sensitive questions.

reports variant-specific coverage and fixed-cohort comparisons.

6.5 Answer generation under matched contexts

Table 5 compares generated answers under the same evidence budgets. The selected ranking reaches 21.67% strict F1, versus 18.33% for original BM25 with recency. The paired difference is 3.33 points, with interval $[-4.84, 11.67]$. Possible and guaranteed support produce identical answer sets on the primary sample.

Their answer sets differ on four diagnostic questions, including three with valid responses under both policies. Earliest and latest date completion yield the same change counts. One valid change recovers the annotated answer. The diagnostic connects context sensitivity to observed answer changes. Appendix G.2 reports exact match, paired intervals, and response-format failures.

7 Related Work

Temporal memory and replacement. Graphiti closes earlier facts through direct contradictory evidence (Rasmussen et al., 2025; Graphiti contributors, 2026). MemStrata applies deterministic supersession to structured keys (Yadav, 2026). ERASE edits stored fact histories as new documents arrive (Li et al., 2025). NuggetIndex attaches source spans, validity intervals, and lifecycle states to atomic records (Zerhoubi et al., 2026). It distinguishes functional and multi-valued relations. We compile direct replacement decisions across uncertain occurrence dates.

Temporal retrieval. Re³ combines fact grouping, conflict filtering, and temporal relevance (Cao et al., 2026). DeFuzzRAG adapts temporal granularity

and filters inferred document scopes by query overlap (Chen et al., 2026). IA-RAG represents intervals and refines uncertain event dates (Wang et al., 2026). TA-RAG retrieves evidence across timelines (Lau et al., 2025). Our compiler preserves every assignment within supplied occurrence bounds. The resulting compiler distinguishes possible support from guaranteed support within a window. The latter uses a different quantifier order from overlap with certainly active points.

Uncertain temporal data. Anselma et al. (2013) study valid-time indeterminacy and compact representations. Fan et al. (2012) study data currency without reliable timestamps. Amarilli et al. (2017b) study possible ordered answers and certainty of an entire query result. Our model restricts uncertainty to independent occurrence bounds with a fixed replacement predicate. Within this model, two direct witnesses answer all support queries.

Uncertain ranked answers. Hua et al. (2008) study top- k membership probabilities under uncertain record membership. Amarilli et al. (2017a) study unknown ranking values constrained by a partial order. Feng et al. (2023) bound possible and certain results for ranking and SQL window queries. We keep relevance scores fixed and derive uncertain eligibility from direct temporal replacement. This structure permits linear stability certification despite NP-hard possible context membership.

Temporal evaluation. TimeQA supplies natural passages and questions about dated histories (Chen et al., 2021). TempLAMA evaluates factual answers at yearly resolution (Dhingra et al., 2022). StreamingQA and FreshLLMs study changing information access (Liška et al., 2022; Vu et al., 2024). FRESCO distinguishes obsolete evidence from current but insufficient passages (An et al., 2026). Our evaluation distinguishes annual observations from occurrence bounds and checks retrieval against original source spans.

8 Conclusion

Two temporal witnesses make date uncertainty actionable at the point of evidence selection. The retriever returns a stable context, or explicit timelines and support decisions that explain why its evidence can change.

Limitations

The guarantees use independent occurrence bounds and fixed directed replacement decisions. Dependencies such as a known succession order can restrict the allowed timelines. The product model then provides conservative support bounds for those dependencies. Incorrect source interpretations can also produce incorrect temporal conclusions. The certificates establish exact decisions for the supplied model, rather than the truth of extracted claims.

The query predicate asks whether evidence applies somewhere within a period. Questions requiring uninterrupted validity or a new appointment within that period need different predicates. Compact certificates answer support and stability queries. Replaying complete contexts also retains the accepted graph. The hardness result treats the context budget as an input. It does not establish hardness for a fixed budget of five.

The adapter models 1.49% of development units and 1.36% of test units. The final evaluation primarily exercises uncertain starts and explicit ends. All recovered cross-claim pairs occur in calibration articles. A recorded model-assisted audit accepts 54 of 60 sampled event interpretations. Eight accepted events omit explicit paragraph-level ends; five further endpoint interpretations remain uncertain. Unknown fallback text remains available without certified temporal support. These findings distinguish source interpretation from exact reasoning over its output.

The reader study uses one small quantized model and a fixed 60-question sample. Its separate eleven-question diagnostic retains an earlier cohort selected for certified context disagreement. TimeQA anchors can be aliases, headings, or mentions outside the relevant period. Strict annotated-answer metrics differ from expert judgments of semantic correctness. Larger readers and human source annotations would strengthen the empirical assessment.

The evaluated questions have previously inspected outcomes. Source-audit findings guide the final adapter revision, which remains frozen throughout the reported evaluation. The earlier TimeQA human and template test tracks overlap. TemplAMA annual labels can combine concurrent values with within-year succession. The matched Graphiti comparison isolates its temporal policy kernel under supplied candidates. It does not eval-

uate Graphiti’s semantic extraction model. The release preserves source licenses, raw results, and each experiment’s scope.

Ethics Statement

The experiments use published historical passages and benchmark questions. The system preserves original source spans for inspection. Incorrect source interpretations can misstate a person’s role or tenure. The certificates describe uncertainty in the supplied records and do not establish historical truth. The release preserves dataset attribution and model licenses. Source assessments use model-assisted review and are identified as such.

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A.1 Point support

Lemma 1 (Exact point support). *For every date t ,*

$$P_{[t,t]}(c) \iff \ell_c \leq t < p_c, \quad (9)$$

$$H_{[t,t]}(c) \iff u_c \leq t \wedge (t \leq \ell_c \vee t < r_c). \quad (10)$$

Proof. If $t < \ell_c$, no assignment makes c active. If $t \geq p_c$, a witness satisfies $\ell_d > u_c$ and $u_d \leq t$. It occurs strictly after c and by t in every assignment. Thus possible support requires $\ell_c \leq t < p_c$.

Conversely, assume these inequalities. Choose $x_c = \min(u_c, t)$. For each accepted witness, choose $x_d \leq x_c$ when its bounds permit that choice. Otherwise, $\ell_d > x_c$. If $x_c = t$, this witness occurs after t . If $x_c = u_c < t$, the definition of p_c gives $u_d > t$. Choose the witness after t in this case. Independent date bounds permit all these choices simultaneously. No witness then retires c by t .

For guaranteed point support, every assignment has $x_c \leq t$ exactly when $u_c \leq t$. Assume this condition. A witness can retire c by t exactly when some assignment satisfies $x_c < x_d \leq t$. Choose $x_c = \ell_c$. A suitable witness date exists exactly when $t > \ell_c$, $\ell_d \leq t$, and $u_d > \ell_c$. The minimum eligible ℓ_d is r_c . Possible retirement therefore requires $t > \ell_c$ and $r_c \leq t$. Its complement gives the stated condition. $\square$

The equality term is necessary. Take $[\ell_c, u_c] = [0, 0]$ and $[\ell_d, u_d] = [0, 1]$ with $G(d, c)$ true. Claim c is certainly active at zero. A strictly later replacement cannot have occurred by zero. It is not certainly active at any later date. An ordinary half-open interval cannot represent this singleton. The implementation retains the exact comparisons from the lemma.

A.2 Window support

Proof of Theorem 1. Possible point support occupies $[\ell_c, p_c)$ by Lemma 1. This set intersects $[a, b]$ exactly when $\ell_c \leq b$ and $a < p_c$. The two existential quantifiers commute, proving Equation 5.

For guaranteed support, first suppose $u_c > b$. The assignment $x_c = u_c$ gives no support within the window. Thus $u_c \leq b$ is necessary. Now assume that inequality. Every assignment with $x_c \geq a$ supplies support at x_c itself. For an earlier occurrence, support fails exactly when a direct witness retires c by a . Such an assignment exists exactly when some witness satisfies $x_c < x_d \leq a$. The preceding lemma shows this is equivalent to

$a > \ell_c$ and $r_c \leq a$. Taking its complement proves Equation 6. $\square$

For each world, support means existence of a suitable time. The suitable time can differ across worlds. Consequently, a window can have guaranteed support even when no point has certain support. This distinction remains after exact text matching.

A.3 Sharpness and refinement

The possible filter retains precisely the union of world-specific support sets. Any smaller filter omits a supported claim in some world. The guaranteed filter retains precisely their intersection. Any larger filter includes an unsupported claim in some world. These statements establish the claimed minimal and maximal filters. They concern each retained claim's support, rather than one jointly optimal context.

Narrowing date bounds replaces Ω with a subset $\Omega' \subseteq \Omega$. Existential support can only disappear when its assignment set shrinks. Universal support can only appear. This proves refinement monotonicity without assumptions about gate transitivity. For correlated dates, a product approximation instead gives conservative bounds. It can enlarge possible support and reduce guaranteed support.

A.4 Local gate influence

Suppose gates G and H differ on directed pairs with target set U . Every claim outside U uses identical terms in both boundary minima. Its support status therefore remains unchanged for every query window. Thus either mask difference lies within U , where $|U| \leq h$.

Fix a complete candidate ranking and deterministic tie handling. Changing one eligibility bit alters at most one selected item and one replacement. Apply this argument successively to the targets in U . The resulting context difference is at most $2|U|$. Each context also contains at most k items. The difference is therefore at most $2 \min(k, |U|) \leq 2 \min(k, h)$.

A directed gate edit differs from a symmetric matching edit. A symmetric edit can change both directions and affect two target claims. Overlapping date bounds do not define one fixed older endpoint. A date edit can also affect several target certificates. The local bound does not treat either edit as one directed decision.

A.5 Exact dates and compilation cost

When $\ell_c = u_c = x_c$ for every claim, both boundary minima select the earliest strictly later direct witness. The possible and guaranteed masks then coincide for every window. For a point query, they equal the exact-date earliest-witness policy.

The reference compiler evaluates all $n(n-1)$ candidate directed gates. The certificates require this scan or a proved complete candidate index. Two stored witnesses alone cannot verify that no omitted witness changes a boundary. If accepted pairs are already available, two minima require $O(n+m)$ time for m pairs. The output contains at most $2n$ witness records. Every point or window decision takes $O(1)$ time per claim. A full mask takes $O(n)$ time. A complete text index can reduce gate evaluations without changing these results. No subquadratic guarantee is asserted for candidate discovery.

The worked certificate in Section 4 proves worst-case witness minimality.

B Source and Evaluation Details

B.1 TempLAMA source reconstruction

We recover the complete official test release and a public alias mirror. Both contain the same 34,963 records and the same answer identifiers. The normalizer verifies record IDs, query text, first answers, and full answer sets. The source manifest stores byte hashes for both inputs. It preserves all 45,866 answer observations.

A source key combines the subject identifier and relation identifier. The revised evaluation reserves all 300 previously used keys for legacy development. A deterministic key split assigns 764 additional keys to development and 2,993 to test. No key appears in two partitions. The test contains 25,752 yearly records and 33,908 answer observations. All thresholds and baseline choices use development records.

A year is a benchmark observation bucket. It is not a publication timestamp or an independently observed update date. Multiple answers can reflect within-year succession or concurrent values. An answer absent from a later label set is not independently verified as retired. The source audit consequently reports changes in label sets. It does not convert these changes into new occurrence bounds.

Textual names do not always identify one source entity. Six parsed slots merge 13 source entities. Fifty identical question-year groups cover 105

source rows. Sixty-one of those rows occur in the held-out test partition. The evaluation keeps these cases. It does not expose hidden entity identifiers to the text parser.

B.2 TimeQA provenance and positive labels

The source repository is pinned to commit 38b0598; the manifest records its full identifier. The release includes immutable download URLs, SHA256 hashes, and Git blob checks. The normalizer preserves each original paragraph and its character offsets. Repeated relation histories reuse the same paragraph identifier. The test corpus contains 33,493 distinct paragraphs.

The test annotations contain 384 empty answers with zero-length spans. The human subset contains 159 such annotations. The positive retrieval tracks exclude these placeholders. They retain 2,613 template-rendered questions and 830 original human questions. The human track is a subset of the larger test collection. It is not an independent second benchmark.

Eight articles occur in both the published development and test collections. Removing them before development produces 2,651 nonempty development questions. The natural-passage controls have fixed rules and no fitted parameters. The source audit records all removals and overlapping article paths.

A matched string span identifies the benchmark’s retrieval anchor. Some anchors appear in headings or in passages about another period. Aliases can also create several labels for one entity. The evaluation therefore names its metrics annotated passage hit and annotated span coverage. Entailment requires a separate source-support judgment.

B.3 Source-grounded examples

The Chicago Stadium paragraph gives two overlapping tenancies. The Blackhawks’ stated period is 1929–1994. The Bulls’ stated period is 1967–1994. Both cover the question’s period, 1967–1988. This example supports coexistence directly from the passage.

The News Quiz paragraph describes several hosts between 1977 and 1995. It includes Barry Norman, Barry Took, and Simon Hoggart. Barry Took returns after an intervening host. This example illustrates range coverage and recurrence. Its multiple answers do not establish simultaneous hosts.

A Ruud Lubbers paragraph mixes month-level and day-level employment dates. The associated question uses year-level bounds. Other passages use expressions such as *before 1516* and *around 1713*. These examples motivate preserving source precision. They do not supply numerical uncertainty widths by themselves. The release records their original text and exact offsets.

C Archived Results and Source Support

The supplement preserves previous generations, metrics, and source files. These records use a selected-answer protocol. Their answer metric checks normalized target containment. It does not require the complete answer set. The new source-only reader runs use a separate complete-answer protocol.

The previous TempLAMA builder kept the first listed answer. Its 300 keys contain 447 changes in that selected answer. In 153 cases, the previous answer remains in the new full answer set. The revised scorer instead accepts every official answer identifier. The annual controls use this complete-source protocol.

The archived extractive reader accesses an annotated value field. The archived historical evaluator also uses a gold anchor. Those outputs remain reproducible as historical records. They are separate from the source-only controls in the main body.

The RealProse source audit identifies 17 excerpts without their assigned subject-role statement. It identifies 33 excerpts without the assigned start year. One excerpt gives a different start year. These findings concern the supplied passages, rather than the underlying facts. TimeQA provides the primary natural-text retrieval source in this revision.

D Reproduction

The release includes the paper sources, vector figures, compilers, corpus normalizers, predictions, and result cards. A verification entry point reruns the compiler checks and reconstructs the paper tables. Longer entry points rerun full-source and natural-passage retrieval. Input and prediction hashes identify the exact evaluated artifacts. The Graphiti policy adapter includes the inspected upstream code segments and their license. It isolates temporal invalidation under supplied candidates. It does not reproduce Graphiti’s semantic extraction model.

E Explicit Lifetimes and Fixed Fallbacks

Let claim c start at $x_c \in [\ell_c, u_c]$ and end at $z_c \in [v_c, w_c]$. Assume $u_c < v_c$ and independent start and end dates. Its explicit lifetime is $[x_c, z_c]$, additionally shortened by accepted direct replacements. Introduce a hidden event e_c with bounds $[v_c, w_c]$ and the pair $e_c \rightarrow c$. Every world has $x_c < z_c$. The hidden event retires c exactly when the explicit end occurs. It therefore preserves activity in every world. The support boundaries update to

$$p'_c = \min(p_c, w_c), \quad r'_c = \min(r_c, v_c).$$

The same support formulas apply, with at most two witness records per claim. Hidden end events never enter a reader’s context.

The strict separation condition is necessary for this independent-date reduction. For start bounds $[0, 2]$ and end bounds $[1, 3]$, the product permits start 2 and end 1. A valid lifetime excludes that assignment, but direct replacement ignores an earlier event. The adapter therefore leaves overlapping start–end bounds unmodeled. It also retains approximate dates without justified finite bounds as unmodeled text.

Let F contain fixed fallback items and S_x contain the modeled support set. The operational mask is $S_x \cup F$. Its union over worlds is $P \cup F$, and its intersection is $H \cup F$. The context test therefore compares $\text{Top}_k(P \cup F)$ with $\text{Top}_k(H \cup F)$. Fallback eligibility records a policy choice and never receives the guaranteed-support tag.

Additional source constraints can restrict the independent date rectangle. Possible support then remains an outer bound, while guaranteed support remains an inner bound. Equal endpoint contexts still certify stability under the extra constraints. Different endpoint contexts need not establish instability within that smaller family. Counterworld exactness uses the stated independent-date model.

F Constructive Context Uncertainty

Fix a query $Q = [a, b]$, a complete strict ranking, and a budget k . Let P and H be the exact possible and guaranteed support sets. Let F contain any source items retained under a fixed fallback policy. Fallback items have unknown temporal support. They enter every operational retrieval mask. Define

$$U = \text{Top}_k(P \cup F) \setminus (H \cup F).$$

The exact context theorem makes U empty precisely when the context is stable.

Construct two witnessing timelines. Choose $c \in U$. Fewer than k items in $P \cup F$ outrank c . Every world supporting c must therefore select it. Set $t = \max(a, \ell_c)$ and $x_c = \min(u_c, t)$. Possible support gives $t \leq b$ and $t < p_c$. For every nonpivot event d , assign

$$x_d = \begin{cases} \ell_d, & \ell_d \leq x_c, \\ u_d, & \ell_d > x_c. \end{cases}$$

The first case places d no later than c . For every accepted witness, the second case places d after t . Indeed, if $x_c = t$, then $u_d \geq \ell_d > t$. Otherwise, $x_c = u_c < t$ and $\ell_d > u_c$. The definition of p_c then gives $u_d \geq p_c > t$. This rule also makes valid choices for all events outside the incoming witness set. Those choices cannot affect whether c is active. The assignments respect all independent date intervals. Claim c is active at t , so this world selects it.

For a second world, first consider $u_c > b$. Set $x_c = u_c$ and every other date to its lower bound. The claim starts after the query and cannot enter its context. Otherwise, failed guaranteed support gives $a > \ell_c$ and $r_c \leq a$. Choose the stored witness d for r_c . Set $x_c = \ell_c$ and $x_d = \min(u_d, a)$. The witness conditions give $\ell_d \leq x_d \leq u_d$ and $\ell_c < x_d \leq a$. Thus d retires c by the query start. Set every remaining date to its lower bound. This world cannot select c . The two contexts differ.

Let n count modeled events and N count ranked excerpts, including fixed fallback items. The construction scans the event bounds and pivot certificate in $O(n)$ time. It returns $O(n)$ date values and needs no retained pair list. The pivot's rank condition already proves that these assignments yield different contexts. Optional replay scans the accepted pairs and ranking in $O(n + m + N)$ time. Explicit end events enter the same construction but remain hidden from the context. Direct context replay retains the accepted pairs and uses $O(n + m + N)$ audit state. The compact $O(n)$ certificate output stores support boundaries, rather than the full graph.

Every current pivot must be resolved. Narrow the date intervals without emptying any interval. Keep all other inputs fixed. Let P' and H' denote the refined support sets. Then $P' \subseteq P$ and $H \subseteq H'$. Take $c \in U$ that remains in P' . It still has

fewer than k possible items above it. Thus $c \in \text{Top}_k(P' \cup F)$. If the refined context is stable, the exact context theorem requires $c \in H' \cup F$. Since $c \notin F$, it must belong to H' . Therefore, each current pivot becomes guaranteed or impossible.

This condition is necessary but not sufficient. For example, rank two claims c, d with independent start bounds $[0, 2]$. Use no replacement pairs, point query 1, and budget 1. Initially, the frontier is $\{c\}$. Refining c to start 2 resolves that pivot but exposes frontier $\{d\}$. The result therefore gives no minimum number of date refinements.

Hardness of the full possible context envelope.

We reduce Set Cover (Karp, 1972) to possible top- k membership. The context budget k is part of the decision input. Take a universe E , sets $S_1, \dots, S_s$, and budget $1 \leq K \leq s$. Create variable claim v_j with date interval $[1, 3]$ for every set. Create $K + 1$ copies of each element claim, all dated exactly 0. Create candidate c dated exactly 0. Accept $v_j \rightarrow e$ for every copy of each $e \in S_j$. There are no other accepted pairs. The gate is bipartite, with one edge layer from variables to element copies.

Rank all variables and element copies above c . Set query time 2 and budget $k = K + 1$. Candidate c has guaranteed support. A variable is active exactly when its date is at most 2. Every active variable retires its associated element copies by the query time. An inactive variable starts after the query and cannot retire any copy by that time.

Given a cover of size at most K , assign its variables date 1 and others date 3. Every element copy is retired. At most K higher-ranked variables remain active, so the context includes c . Conversely, suppose a world includes c . At most K higher-ranked claims are active. The variables dated at most 2 form a collection of at most K sets. If they miss an element, its $K + 1$ active copies exclude c . They must therefore cover E . The construction has polynomial size and proves NP-hardness. It does not establish hardness for a fixed budget such as $k = 5$.

The result also holds when every start occurs by the query time. For this variant, use variable bounds $[0, 2]$, element dates 1, and candidate date 2. Add a clock at date 1, ranked below the candidate, with a pair targeting each variable. Use the same variable-to-element pairs and query time 2. Variables before 1 retire under the clock. Variables at or after 1 remain active, but only those after 1 retire element copies. A cover assigns its variables

2 and all others 0. Conversely, a selected candidate permits at most K active variables. Those after 1 must cover every element, or $K + 1$ active copies exclude the candidate. Variables exactly at 1 add cost without covering an element. This variant has an acyclic gate with two edge layers.

This hardness concerns arbitrary possible context membership. It does not contradict the linear exact stability test. An unstable possible context already supplies a pivot with fewer than k possible items above it. That rank condition makes its supporting world sufficient to demonstrate a context change.

G Source Pipeline and Run Provenance

G.1 Calibration, revision, and source interpretation

Removing eight development–test article overlaps leaves 2,651 eligible development questions. A fixed hash selects 176 questions from 50 articles for the ranking pilot. The initial remainder contains 2,475 questions from 679 articles. Earlier source inspections identify 35 further articles for exclusion before their first validation outcomes. Removing their 127 questions gives the 2,348-question template track. The human track retains 830 positive questions from the published test collection. Both tracks had inspected outcomes before the final source revision. The final evaluation preserves their questions, ranking, budgets, reader settings, parser, and metrics. The release identifies these runs as post-audit revision comparisons.

Initial source calibration covers 86 articles, including 85 with eligible development questions. It combines 50 pilot articles with 39 previously identified source paths, with three overlaps. Ranking selection uses only the 176-question pilot. An independent feature audit changes query metadata and remaps source identifiers without changing selections. Ranking accepts the question string alone. The title inventory comes from the source corpus. Every ranking retains every corpus unit exactly once.

A year maps to January 1 through December 31 on the Gregorian ordinal axis. A month maps to its actual first and last dates. A fully stated day gives a singleton. Tenures keep separate uncertain start and end events. The final adapter requires local subjects, preserves abbreviation boundaries, checks date ownership, and links uniquely resolved paragraph-level ends. Recognized bounded states

with unresolved endings remain fallback text. No answer labels guide these repairs. Twenty-seven source checks pass before the final evaluation. Five inspected calendar-end cases receive explicit endpoints. Six inspected duration cases remain fallback text because their endings cannot enter the independent-bound model. These eleven cases guide development and do not estimate extraction accuracy.

All source words survive the modeled/fallback partition. Independent checks find no overlapping units and no rejected pair endpoints. The development and test pools contain 33,005 and 34,082 units, respectively. Their modeled claim counts are 493 and 464, with 292 and 238 explicit ends. Sixteen paragraph-level end links enter the two pools. The development corpus contains six succession starts and three directed pairs. The test corpus contains two succession starts and no pairs. All three pairs occur in previously inspected calibration articles. The Warburg example uses automatic extraction; its question period and ranking are illustrative. Its selected narrative dates Bing’s appointment to 1955, while another paragraph lists 1954. The example compiles the stated narrative without resolving that cross-paragraph discrepancy.

Earlier source audit. The preceding frozen version was assessed on 60 development claims outside its calibration articles. That model-assisted audit accepted 52 events, rejected five, and marked three uncertain. Eleven accepted starts omitted terminations elsewhere in their paragraphs. Twenty-seven of its 29 explicit-end records had accepted event grounding. The final adapter addresses those findings through the five source-attachment checks described above. The earlier judgments remain unchanged in the release.

Recorded source audit. After the adapter freeze, a fixed hash selects 60 claims from 55 articles. The sample contains twenty development claims and forty test claims. It excludes 277 documented or identifiable exposure articles. The complete membership of a lost earlier audit is unavailable. We cannot rule out prior exposure for every article. Reviewers read complete source paragraphs without benchmark questions, labels, retrieval scores, or reader outputs. They accept 54 core interpretations, reject four, and mark two uncertain. All 408 checked spans match the source characters exactly. Twenty-six of 27 supplied ends are accepted; one is rejected. Core correctness and lifecycle complete-

ness receive separate judgments. Eight accepted events omit explicit paragraph-level ends; five further endings have uncertain applicability. These model-assisted judgments are neither human annotations nor extraction-recall estimates. The samples differ, so the earlier 52/60 and current 54/60 are not paired improvement estimates. No judgment changes the final frozen adapter.

G.2 Reader protocol

The current model is Qwen2.5-3B-Instruct, Q4_K_M, at revision 7dabda4d. The release records the full revision and weight checksum. The runtime is llama.cpp b11435 at commit 43fe9c6. The model uses the Qwen Research License; the runtime uses MIT. Generation weights are excluded from the archive. Pinned download scripts restore the official artifacts.

Prompts contain the visible question, original excerpts, source titles, date bounds, and temporal status. Modeled excerpts also carry source-derived subject and relation labels. Those labels preserve role context for short succession clauses. No extracted answer-value field is added to the prompt metadata. Temperature is zero, seed is 1729, and the output limit is 96 tokens. Prompt caching and JSON grammar enforcement are disabled. A tokenizer check verifies that every request fits the configured context. Resource-only amendments precede completed question-answer outputs and preserve the original freezes and logs. The final runtime uses four threads, a 2,048-token context, and polling disabled. Its batch and microbatch sizes are both 512.

The parser accepts a complete JSON object or a single JSON code fence. It rejects narrative repair, invalid field types, invalid excerpt indices, and truncated outputs. Failures receive empty answer lists and remain in every denominator. Normalization applies Unicode NFKC, case folding, and word-token extraction. It preserves articles and compares deduplicated complete answer sets. Strict set precision, recall, F1, and exact set accuracy use those normalized sets. Token overlap remains a secondary measure.

The primary sample contains 60 questions from 55 articles. Its 600 method conditions reduce to 194 distinct prompts. The diagnostic retains eleven questions selected by the preceding source version without answer labels or generated answers. That version selected different possible and guaranteed contexts, up to twenty-four questions. The revised

Method	Primary		Diagnostic	
	EM	F1	EM	F1
Original BM25	16.67	16.67	9.09	9.09
Original BM25 + year	18.33	18.33	9.09	9.09
Title-aware ranking	21.67	21.67	9.09	9.09
+ earliest completion	20.00	20.00	18.18	18.18
+ midpoint completion	20.00	20.00	9.09	9.09
+ latest completion	20.00	20.00	9.09	9.09
+ interval control	20.00	20.00	9.09	9.09
+ possible support	20.00	20.00	9.09	9.09
+ guaranteed support	20.00	20.00	18.18	18.18
Title-aware + year	21.67	21.67	9.09	15.15

Table 6: Reader exact match (EM) and strict answer-set F1 (%). Primary: 60 questions and 600 conditions. Diagnostic: 11 questions and 110 conditions. Unique generation parse failures: 69/194 and 11/40, respectively. Failures remain in both denominators.

contexts still differ on seven of those eleven questions. All eleven remain evaluated, yielding 110 conditions and forty distinct prompts. They come from nine articles, four shared with the primary sample. The cohorts have no question overlap and no shared requests. Of 234 current requests, 166 exactly match archived requests under the same model and execution settings. The other 68 requests receive new generations. Reuse checks include full messages, output limits, seed, stop conditions, model hashes, and runtime settings. Full request hashes, raw responses, timing, and per-request provenance support independent verification.

Complete reader outcomes. Table 6 reports both cohorts separately. The diagnostic guaranteed-minus-possible F1 difference is 9.09 points, with article-cluster interval [0.00, 30.00]. All 45 paired method contrasts remain in each cohort’s saved summary. The primary temporal contrasts change no answer sets, including all 37 pairs with valid responses on both sides. The diagnostic possible-guaranteed contrast changes three answer sets among seven pairs with two valid responses. One further change involves a response-format failure. Earliest-latest completion gives the same counts and changed questions. Only the Maher change increases strict F1. The other valid changes concern Atlético Madrid and Helen Clark; both retain zero strict F1. This diagnostic retains its earlier selection while evaluating the revised source contexts.

Across 234 distinct current requests, eighty responses fail the frozen parser. These comprise 58 JSON-format failures, four invalid citation lists, and eighteen length-limit responses. Every failed

response remains scored with an empty answer set. The primary ranking comparison changes twenty answer sets against original BM25 with recency. Five changes occur among 24 pairs with two valid responses; fifteen involve a parsing failure. Separate descriptive audits report this breakdown for every method pair without changing the frozen scores. The release retains 238 earlier executions and 68 new executions, totaling 306 archived executions across both versions.

G.3 Ranking portability and selection cost

A secondary analysis applies the certificate to all six ranking variants considered during calibration. The revised protocol precedes its mechanism analysis and preserves the six variants from the earlier ranking study. It reads no answer labels and does not replace the selected main ranking. Each variant has its own modeled-context denominator. The release also reports fixed main and union cohorts for direct comparisons. The lazy selector agrees with full-mask evaluation for every parsed query. Temporal-test counts include possible and guaranteed predicate calls. Fallback excerpts require no predicate call. The timing microbenchmarks exclude extraction, compilation, relevance ranking, rendering, and answer generation. Concurrent CPU workloads affect their absolute timings. The main text therefore reports operation counts.

G.4 Research assistance and verification

AI assistants contributed to code development, theoretical derivations, manuscript drafting, figure construction, and citation checking. The source interpretation audits are model-assisted judgments. The release contains independent checks, complete execution records, and a submission-form disclosure covering these uses.

G.5 Earlier reader and extraction calibration

Earlier Qwen2.5-1.5B-Instruct runs remain available under their original frozen settings. A 24-question pilot requires 52 generations for 192 conditions. A nine-question diagnostic adds 22 generations and reuses six. The cohorts overlap on two questions, giving 31 distinct questions and 74 actual generations. All methods reach 31.94% strict set F1 on the pilot. The earlier diagnostic changes four answer sets between earliest and latest completion, without improving aggregate strict F1.

Two exploratory learned extractors each process six calibration paragraphs before stopping. The

first returns 41 records, all missing required source quotes. A schema-constrained variant returns 46 records; 45 fail literal grounding. The remaining record has an incorrect holder–date association. Neither extractor supplies claims to the reported confirmation runs. Partial manifests retain all completed outputs, settings, and stopping decisions.

An earlier deterministic reproduction correction sorts BM25 token accumulation and stops after five accepted units. Every one of 20,192 archived method contexts remains unchanged. The original runner and equivalence records are preserved.

G.6 Complete annual controls

The annual split and complete-source labels match the archived experiment. Latest-value selection retains 99.89% first-result accuracy but achieves 86.37% complete answer recall. Latest-batch selection reaches 99.98% recall. The suppression difference is 13.63 points, with key-cluster interval [13.06, 14.21]. Graphiti’s matched policy kernel and direct-first invalidation produce identical end-points across 45,866 observations. These retained results require no repeated model inference.

G.7 Earlier natural paragraph controls

These paragraph runs use the published test passages and positive annotations. Their human-question results were inspected before the source-unit experiments. Paragraph and source-unit runs have different retrieval units and denominators.

Ranking	Template stable/modelled	Human stable/modelled	Template tests	Human tests
Global BM25	285/354	86/102	2.64	2.34
Global BM25 + year	443/534	138/164	2.72	2.39
Title BM25	241/298	76/86	2.66	2.41
Title-aware	189/243	56/66	2.63	2.44
Title BM25 + year	327/396	109/126	2.67	2.42
Title-aware + year	281/343	102/120	2.56	2.40

Table 7: Secondary certificate analysis across all six prespecified rankings. Each stability denominator contains modeled evidence under that ranking. Tests count temporal predicate calls within that modeled cohort.

Source-label property	Count	Rate (95% CI)
Multiple answers	6,498/25,752	25.2 [24.2, 26.2]
Old first retained	1,456/3,980	36.6 [35.1, 38.0]
Any answer retained	6,050/6,961	86.9 [86.0, 87.8]

Table 8: Answer-set properties on held-out source keys. The last two rows condition on first-answer changes and complete-set changes, respectively. Rates and intervals are percentages.

Policy	H@1 ↑	R@5 ↑	S@5 ↓
BM25	64.64	80.84	45.89
BM25 + recent ties	84.04	90.10	40.50
Age weighting	99.81	99.98	24.07
Date-first ranking	99.81	99.98	0.00
Original predicate	99.37	99.77	0.77
Graphiti kernel	99.78	99.93	0.00
Query latest-value	99.89	86.37	0.00
Query latest-batch	99.89	99.98	0.00

Table 9: Annual source-label retrieval on 25,752 held-out questions. All values are percentages. H@1: first-result accuracy; R@5: answer recall; S@5: target-label stale exposure.

Policy	Human		Templates	
	H@5	R@5	H@5	R@5
BM25	45.66	45.04	32.49	31.75
Latest year	54.70	54.02	51.86	50.89
Year envelope	49.16	48.33	34.44	33.67
Explicit range	44.10	43.53	32.11	31.39
Range support	44.58	44.00	38.50	37.63

Table 10: TimeQA paragraph retrieval percentages. H@5 measures annotated-passage coverage; R@5 measures annotated-span recall. The tracks contain 830 human questions and 2,613 template questions.